SPATIAL ANALYSIS OF SCHOOL ACCESSIBILITY IN JOS, NIGERIA USING GIS-BASED COST DISTANCE MODELING

Wisdom Akpabio

Keywords: *Accessibility, GIS, Cost Distance, Education Planning, Nigeria*

ABSTRACT

This study evaluates school accessibility in Jos, Plateau State, Nigeria, using an integrated terrain–road cost-distance model. By incorporating slope-derived impedance and road network characteristics, a realistic travel-time surface was generated and classified into accessibility zones. Zonal statistics were applied to assess the distribution of child population across these zones. Results reveal a highly structured spatial pattern, with optimal accessibility concentrated in urban wards such as Tudun Wada, Gangare, Naraguta, and Angwan Rogo, where dense infrastructure supports efficient movement. In contrast, peripheral wards—including Jarawa Kogi A, Godong, Nakwang, Zandi, and Fadan Fursum—exhibit significantly higher travel times and form distinct clusters of reduced accessibility. A critical threshold is observed at approximately 30 minutes, beyond which accessibility declines sharply.

Population analysis shows that most children reside within highly accessible areas, indicating strong alignment between population and service provision. However, small populations in peripheral wards experience severe accessibility constraints, highlighting a trade-off between efficiency and spatial equity. The findings underscore the central role of road connectivity in shaping accessibility and provide a spatial framework for targeted educational planning.

1. INTRODUCTION

Access to education is widely recognized as a fundamental driver of social development, human capital formation, and long-term economic growth. Within the framework of the Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education), ensuring equitable access to educational facilities remains a priority for governments and development agencies. However, in many parts of the developing world, access to education is not solely determined by the presence of schools, but by the ability of individuals to physically reach them within a reasonable time.

Spatial accessibility, defined as the ease with which services can be reached across space, is a critical but often underexamined dimension of educational equity. Traditional approaches to accessibility assessment have frequently relied on Euclidean distance or simplified proximity measures, which assume uniform travel conditions across space. Such approaches fail to account for real-world constraints, including terrain variability, road network structure, and differences in travel impedance. As a result, they often produce overly optimistic representations of access,

particularly in regions characterized by complex topography and uneven infrastructure distribution.

Geographic Information Systems (GIS) provide a powerful framework for overcoming these limitations through cost-based modeling approaches. By integrating multiple spatial factors—such as elevation, slope, and transportation networks—GIS enables the construction of realistic travel-time surfaces that more accurately reflect how people move through space. Cost distance analysis, in particular, has emerged as a robust method for modeling accessibility, as it accounts for variations in movement difficulty across the landscape.

Jos, located in Plateau State, Nigeria, presents a compelling case for such an analysis. The region is characterized by a combination of undulating terrain, expanding urban settlements, and heterogeneous infrastructure distribution. These factors interact to produce complex patterns of accessibility that cannot be adequately captured using simple distance-based methods. Despite the presence of educational facilities, disparities in access are likely to persist, particularly in peri-urban and fringe areas.

This study addresses these challenges by developing an integrated terrain–road cost surface to model travel time to schools across Jos. By combining terrain-induced impedance with road network friction, the study produces a realistic representation of accessibility and extends the analysis to quantify its impact on child population distribution. In doing so, it contributes to a more nuanced understanding of spatial inequality in access to education and provides a framework for evidence-based planning and intervention.

2. PROBLEM STATEMENT

Despite ongoing investments in educational infrastructure, disparities in access to schools remain a persistent challenge in many urban and peri-urban regions of Nigeria. In Jos, these disparities are not merely a function of the spatial distribution of schools but are shaped by a complex interplay of terrain, transportation networks, and settlement patterns.Conventional planning approaches often assume that proximity to schools equates to accessibility. However, this assumption is fundamentally flawed in environments where physical geography and infrastructure significantly influence mobility. In areas with steep terrain or limited road connectivity, short geographic distances can translate into disproportionately high travel times, effectively reducing access to educational services. Conversely, well-connected areas may exhibit high accessibility despite being located farther from schools.

The absence of a comprehensive, spatially explicit model that integrates these factors limits the ability of planners and policymakers to accurately identify underserved areas. Without such a model, interventions may be misdirected—either by prioritizing areas that are already well-served or by overlooking populations that face genuine accessibility barriers.Furthermore, existing analyses rarely extend beyond spatial patterns to quantify the population-level implications of accessibility disparities. As a result, there is limited understanding of how many children are affected by poor access, where they are located, and the extent to which accessibility constraints impact educational equity.

This study addresses these gaps by developing a GIS-based cost distance model that integrates terrain and road network characteristics to simulate real-world travel conditions. It further extends the analysis by linking accessibility outcomes to population distribution, thereby providing a more comprehensive and policy-relevant assessment of educational access in Jos.

3. OBJECTIVES

3.1 General Objective

The primary objective of this study is to develop and apply a geospatial modeling framework to assess school accessibility in Jos by integrating terrain and transportation factors into a unified travel-time analysis.

3.2 Specific Objectives

To achieve this overarching goal, the study is guided by the following specific objectives:

1. **To derive a terrain-based travel impedance surface** by processing digital elevation data and transforming slope characteristics into time-based movement costs.

2. **To model road network friction** by classifying road types, assigning travel speeds, and converting these into raster-based travel-time representations.

3. **To integrate terrain and road-based impedance surfaces** into a single cost surface that reflects realistic movement conditions across the study area.

4. **To compute travel time to the nearest school** using cost distance analysis, thereby generating a continuous accessibility surface.

5. **To classify accessibility into meaningful categories** based on travel time thresholds that reflect varying levels of service adequacy.

6. **To quantify the distribution of child population across accessibility zones** using zonal statistics, thereby linking spatial accessibility to demographic impact.

7. **To interpret spatial patterns of accessibility and identify areas of concern** with respect to educational equity and service provision.

4. STUDY AREA

The study is conducted in Jos, the capital city of Plateau State, located in north-central Nigeria. Jos lies approximately between latitudes 9°50′N and 10°00′N and longitudes 8°50′E and 9°00′E. The city sits on the Jos Plateau, a highland region characterized by elevated terrain, with average altitudes ranging between 1,200 and 1,400 meters above sea level.

4.1 Physical Characteristics

The terrain of Jos is notably undulating, with a mix of gentle slopes and more rugged features, particularly toward the outskirts. This topographic variability plays a significant role in influencing movement patterns, as slope directly affects travel speed and effort. The presence of rocky outcrops and dissected landscapes further contributes to spatial heterogeneity in accessibility. The climate of the region is relatively temperate compared to other parts of Nigeria, but seasonal variations—particularly during the rainy season—can affect road conditions and mobility, especially in less developed areas.

4.2 Urban Structure and Settlement Patterns

Jos exhibits a mixed urban–peri-urban structure, with a densely populated core surrounded by expanding residential and semi-rural areas. The central districts, particularly within Jos North, are characterized by:

- High population density
- Concentrated infrastructure
- Dense road networks

In contrast, peripheral areas—especially within Jos South and surrounding wards—are marked by:

- Lower population densities
- Fragmented settlement patterns
- Limited infrastructure coverage

This spatial structure creates a gradient of accessibility, where movement conditions deteriorate as one moves away from the urban core.

4.3 Transportation Infrastructure

The road network in Jos varies significantly in quality and connectivity. Major roads (primary and secondary) provide relatively efficient movement corridors within and between urban areas. However, many peripheral zones rely on tertiary or unpaved roads, which are often less reliable and slower, particularly under adverse weather conditions.

This uneven distribution of transportation infrastructure is a key determinant of accessibility and is central to the modeling approach adopted in this study.

4.4 Relevance of the Study Area

Jos provides an ideal setting for this analysis due to the convergence of several critical factors:

- Complex terrain that affects mobility
- Rapid urban expansion into peri-urban areas

- Uneven distribution of infrastructure
- Growing demand for educational services

These characteristics make it necessary to move beyond simplistic accessibility measures and adopt a more integrated modeling approach.

5. DATA AND MATERIALS

Table 5.1: Data and Materials

Data Type	Description
DEM	Digital Elevation Model (30m resolution)
Road Network	OpenStreetMap-derived road dataset
Schools	Point dataset of school locations
Population Raster	Gridded child population dataset
Administrative Boundaries	Study area boundary shapefile

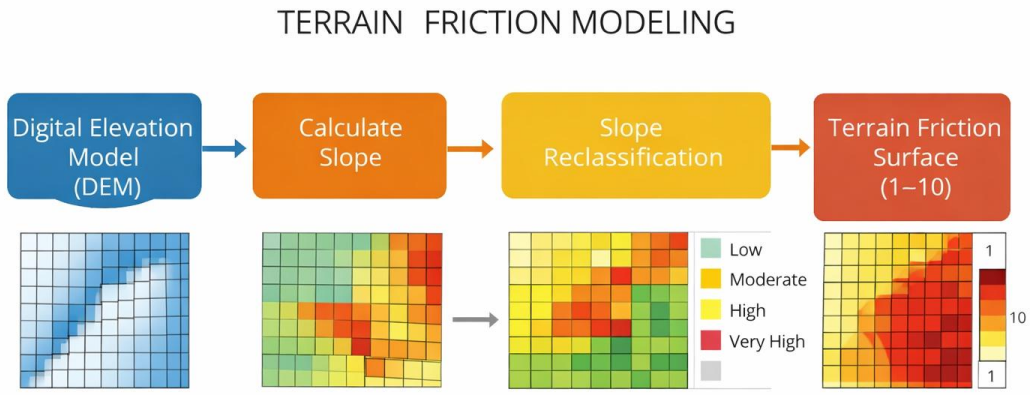
6. METHODOLOGY

This study adopts a multi-stage geospatial modeling framework designed to simulate real-world travel behavior by integrating terrain conditions, transportation networks, and population distribution into a unified accessibility analysis. The methodological approach is structured into sequential processes, each contributing to the development of a realistic travel-time surface and subsequent accessibility assessment.

6.1 Terrain-Based Travel Time Modeling

The first stage of the analysis involved the derivation of a terrain-induced travel impedance surface from a Digital Elevation Model (DEM) with a spatial resolution of 30 meters. Terrain plays a critical role in influencing human mobility, particularly in regions characterized by significant elevation variability such as Jos. Slope was computed from the DEM using standard terrain analysis tools within the GIS environment. The resulting slope raster, expressed in degrees, was then reclassified into discrete categories representing varying levels of movement difficulty. Gentle slopes were assigned lower impedance values, reflecting ease of movement, while steeper slopes were assigned progressively higher impedance values to account for increased physical effort and reduced travel speed.

These reclassified slope values were subsequently converted into a time-based cost surface, representing the estimated time required to traverse each raster cell. This conversion ensures that terrain effects are expressed in the same unit (minutes) as other cost components, thereby enabling seamless integration in later stages. The output of this stage is a terrain travel time surface, which captures the spatial variability of movement constraints imposed solely by topography.

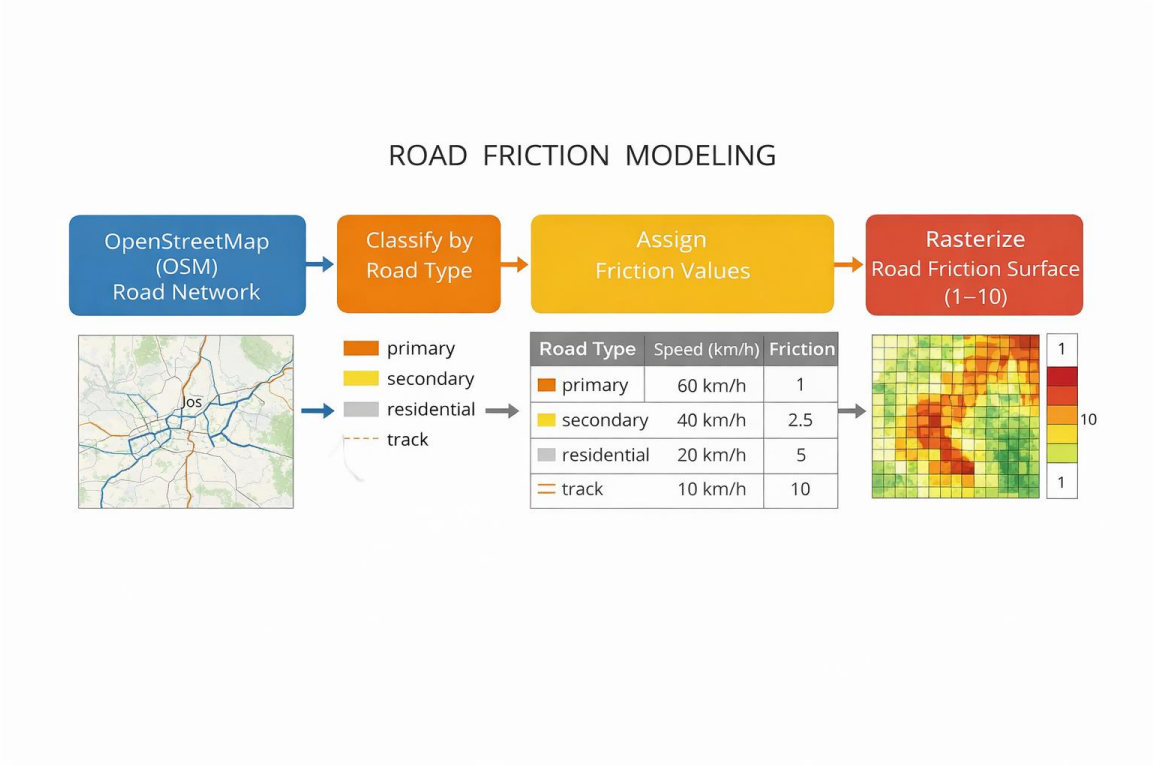


6.2 Road Network Friction Modeling

To incorporate the influence of transportation infrastructure, the road network dataset was processed to generate a complementary friction surface. Unlike terrain, which impedes movement, roads typically facilitate faster travel and thus reduce overall travel time. Each road segment was classified based on its functional hierarchy (e.g., primary, secondary, tertiary, residential), and corresponding travel speeds were assigned based on typical movement conditions. These speed values were then transformed into time-per-cell estimates, ensuring consistency with the terrain-derived surface.

The vector-based road network was subsequently rasterized using the computed friction values. During this process, special attention was given to the treatment of non-road areas. Since rasterization inherently assigns values only to locations where features exist, areas without roads initially resulted in null values. To address this, non-road cells were explicitly assigned high impedance values, ensuring that off-road movement remains possible but comparatively slower.

The result is a continuous road friction raster, where lower values correspond to faster travel along roads and higher values represent slower off-road movement.

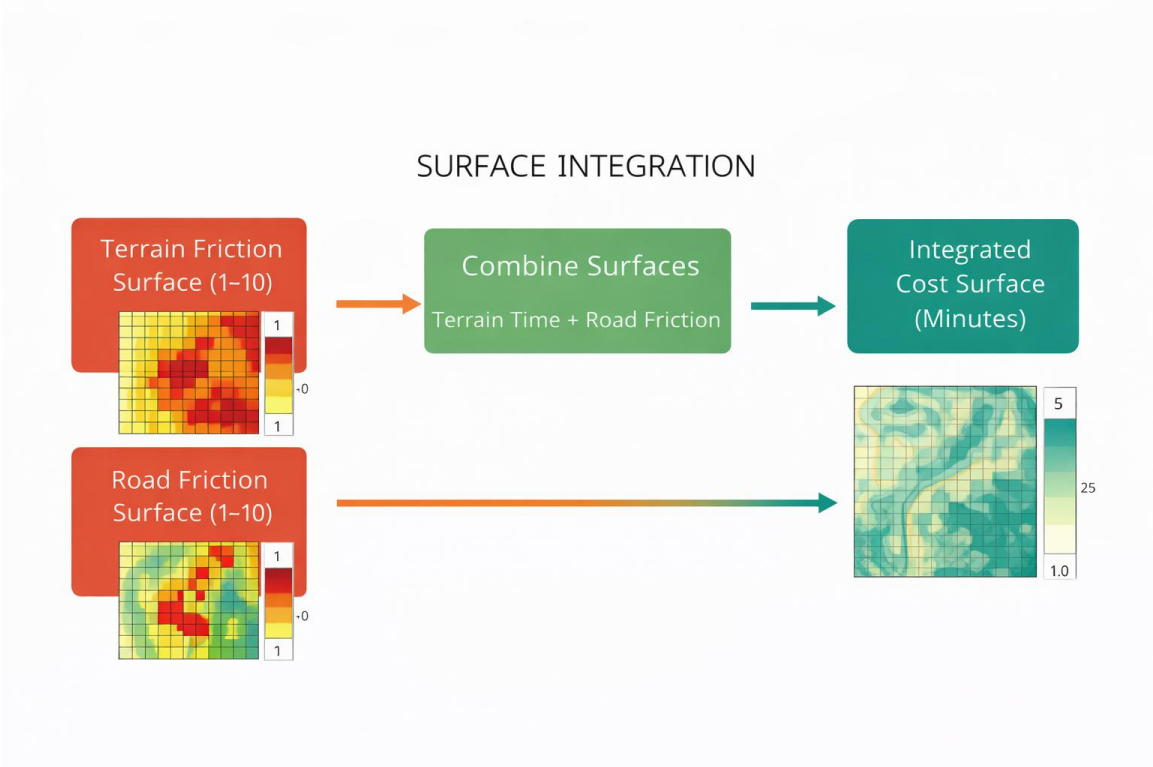


6.3 Integration of Terrain and Road Surfaces

Following the independent generation of terrain and road friction surfaces, both layers were integrated to produce a unified cost surface representing overall travel impedance. This integration was achieved through cell-by-cell raster addition:

$$\textit{Integrated Cost Surface} = \{\textit{Terrain Time}\} + \{\textit{Road Friction}\}$$

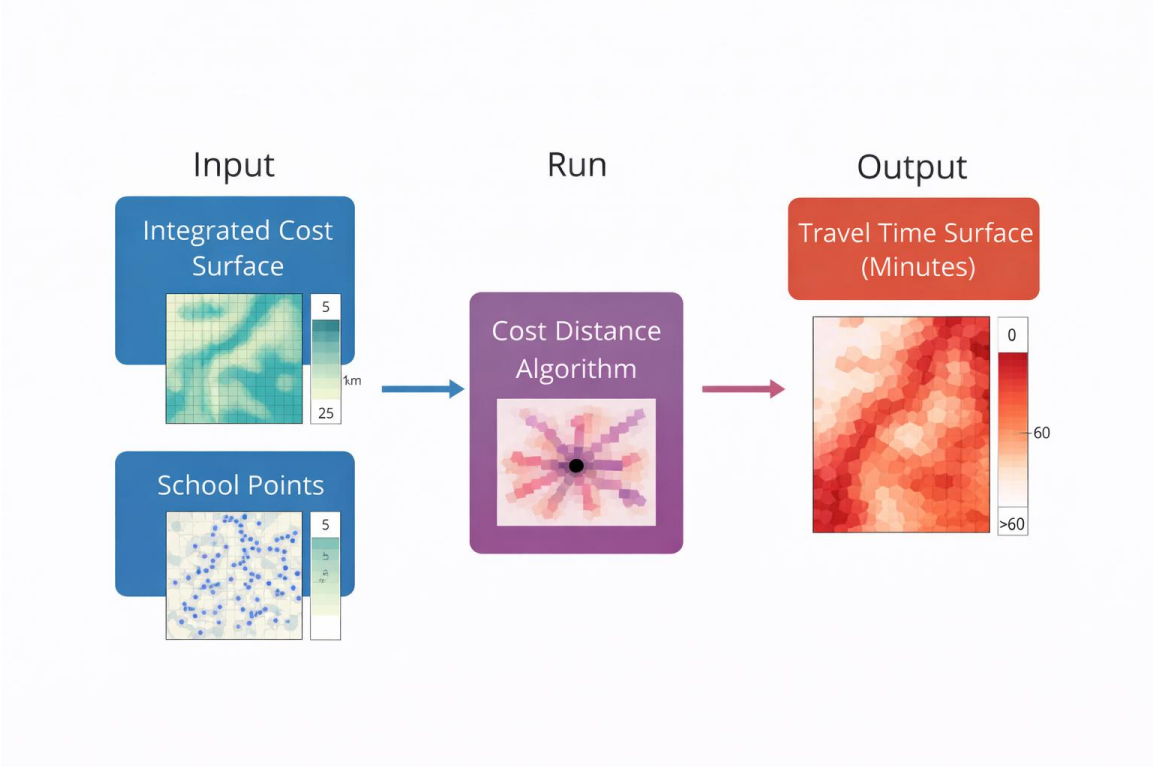
This formulation reflects the combined influence of terrain and infrastructure on movement. In areas where roads are present, lower friction values reduce total travel time, effectively modeling the facilitative role of transportation networks. Conversely, in areas lacking road infrastructure, terrain-induced impedance dominates, resulting in higher travel costs. The resulting integrated cost surface provides a realistic spatial representation of travel difficulty across the study area, expressed uniformly in minutes.



6.4 Cost Distance Analysis

To quantify accessibility, a cost-distance analysis was performed using the integrated cost surface as the impedance layer and school locations as destination points. The cost-distance algorithm computes the least accumulative travel time required to reach the nearest school from every raster cell in the study area. Unlike Euclidean distance, this method accounts for spatial heterogeneity in movement conditions, including terrain variability and road network distribution.

The output is a continuous travel time surface, where each cell value represents the minimum time required to access the nearest school. This surface forms the foundation for subsequent classification and analysis.



6.5 Classification of Accessibility Zones

To facilitate interpretation and policy relevance, the continuous travel time surface was reclassified into discrete accessibility zones based on defined temporal thresholds.

These thresholds were selected to reflect meaningful distinctions in service accessibility:

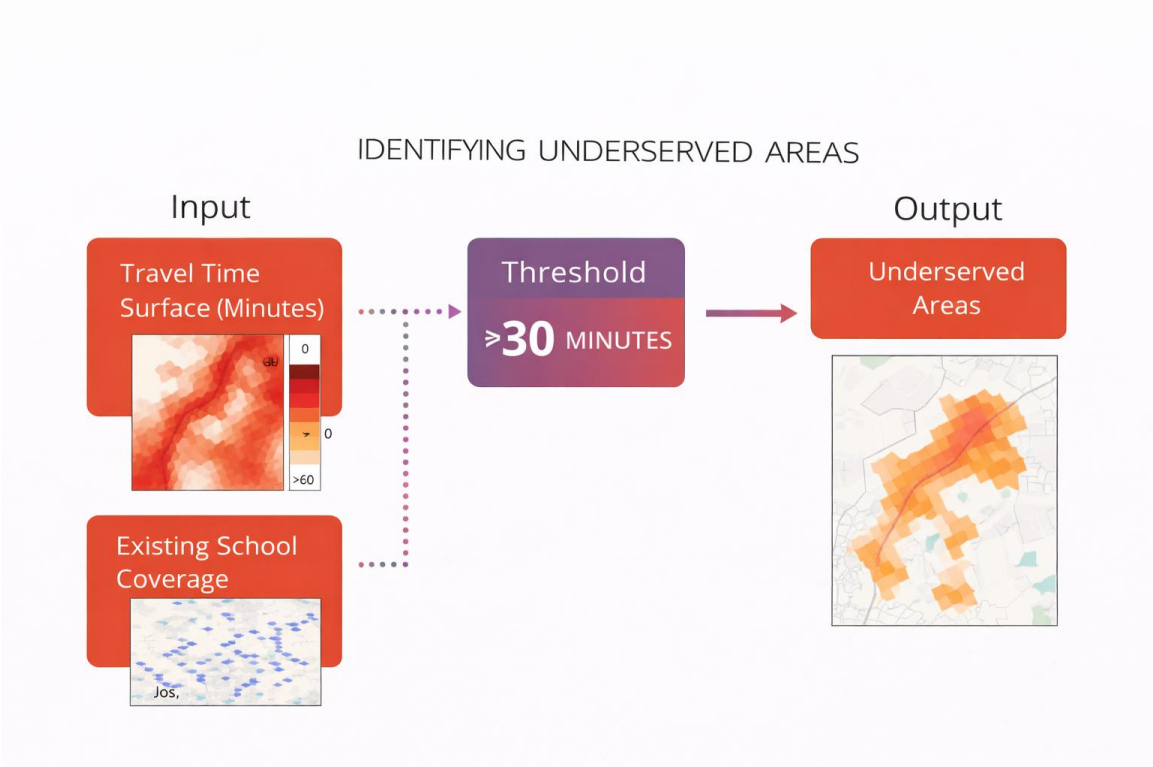
- **0–15 minutes:** Excellent access
- **15–30 minutes:** Good access
- **30–45 minutes:** Moderate access
- **45–60 minutes:** Poor access
- **Greater than 60 minutes:** Critical isolation

The classification process transforms the continuous surface into a categorical map, enabling clearer identification of spatial disparities and facilitating comparative analysis across regions.

6.6 Identification of Underserved Areas

Building upon the classified accessibility zones, underserved areas were identified as regions where travel time exceeds 30 minutes. This threshold reflects a practical limit beyond which access to educational services becomes increasingly constrained.

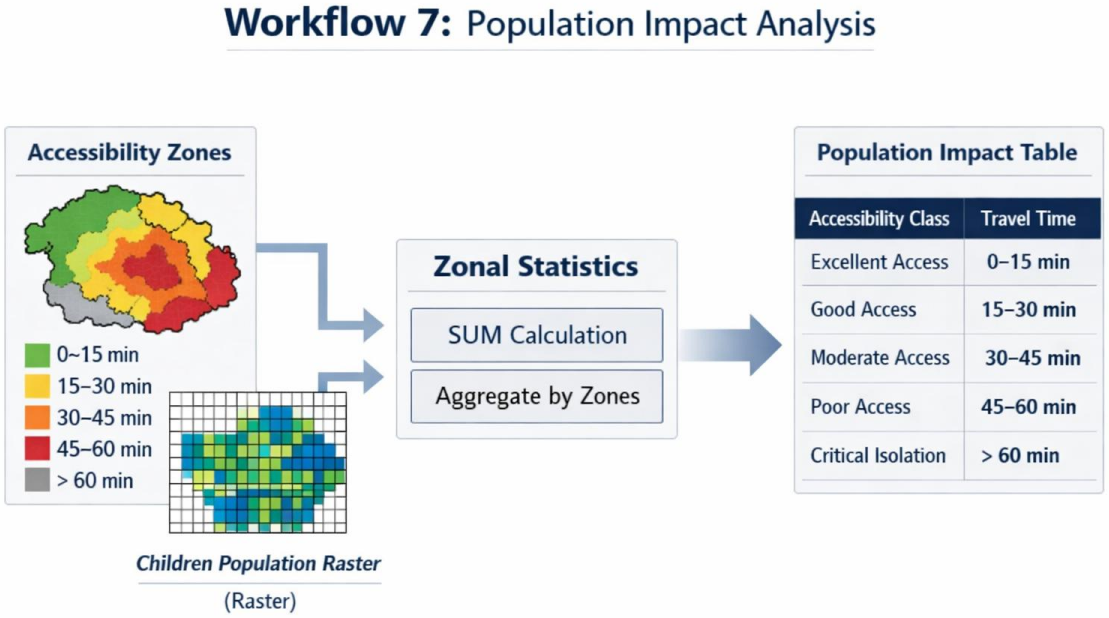
These areas were spatially delineated and analyzed in relation to the distribution of schools and settlement patterns. The identification of underserved zones provides a critical basis for targeted intervention and infrastructure planning.



6.7 Population Impact Assessment Using Zonal Statistics

To quantify the human implications of accessibility disparities, a zonal statistics approach was employed to aggregate population data within each accessibility class. The classified accessibility zones (vector polygons) were used as the zoning layer, while the child population raster served as the input value layer. The sum statistic was computed for each zone, representing the total number of children residing within each accessibility category.

This step translates spatial accessibility into population-level metrics, enabling the identification of not only where access is limited, but also how many individuals are affected.

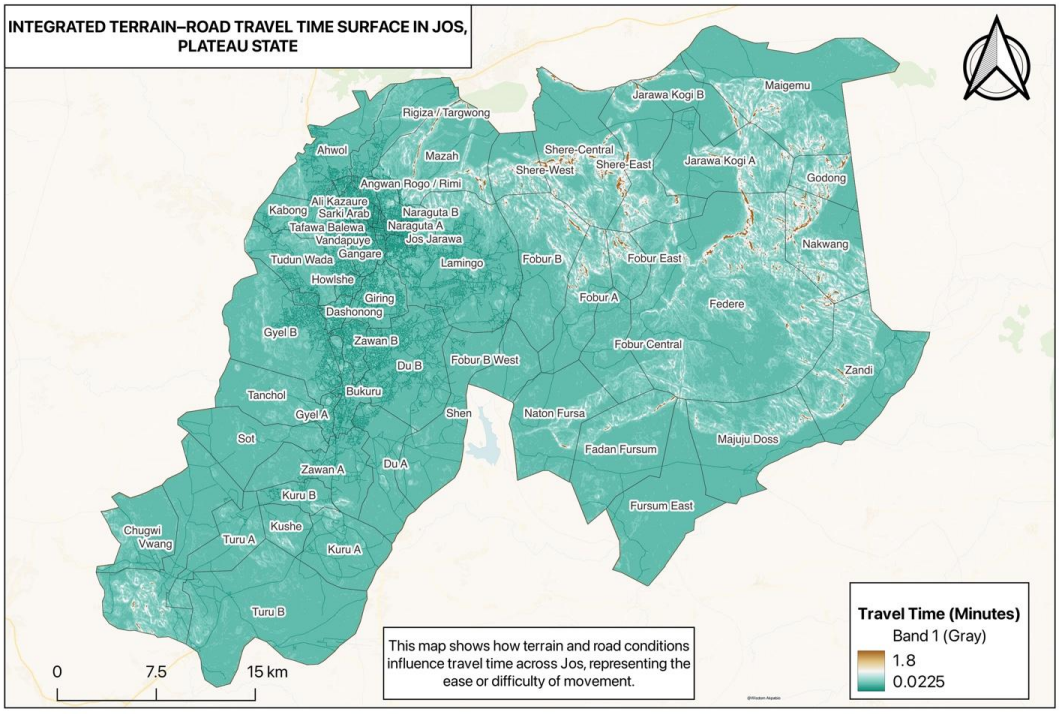


6.8 Synthesis of Analytical Outputs

The outputs from the preceding steps—including the integrated cost surface, accessibility zones, and population statistics—were synthesized to produce a comprehensive understanding of spatial accessibility dynamics within the study area.

Together, these outputs provide both a spatial visualization of accessibility patterns and a quantitative assessment of their impact, forming a robust foundation for evidence-based decision-making.

7. RESULTS AND DISCUSSION

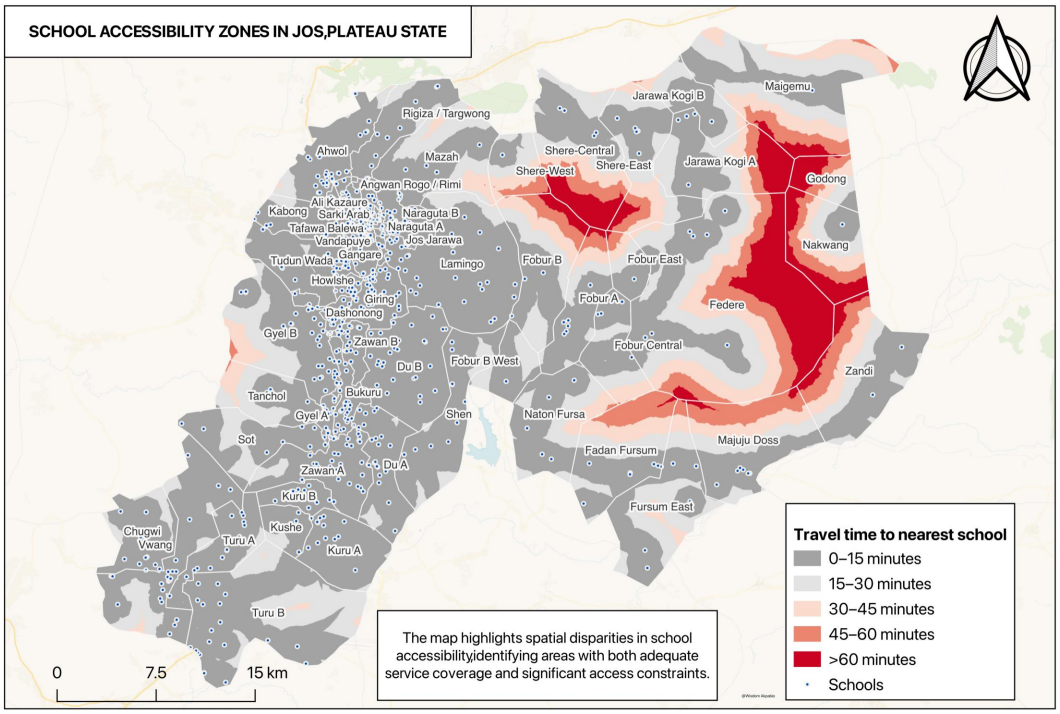


7.1 Integrated Terrain–Road Travel Time Surface

The integrated terrain–road travel time surface provides the foundational representation of movement cost across Jos, revealing how terrain and infrastructure jointly influence accessibility. The spatial pattern is strongly polarized between the urban core and the peripheral regions. Low travel time values dominate the central and western wards, particularly in Tudun Wada, Gangare, Angwan Rogo/Rimi, Naraguta A and B, and Jos Jarawa. These areas are characterized by dense and interconnected road networks, which significantly reduce movement impedance and offset terrain constraints. The resulting surface appears relatively smooth and continuous, indicating efficient mobility and minimal barriers to movement.

In contrast, the eastern sector of the study area, especially Jarawa Kogi A, Godong, Nakwang, and extending toward Zandi, exhibits elevated and spatially continuous travel time values. The surface here is fragmented and intensified, reflecting the combined influence of rugged terrain and sparse infrastructure. These wards form a pronounced high-cost corridor, indicating systemic movement constraints. A similar pattern is observed along the southern fringe, particularly in Fadan Fursum, Naton Fursa, and parts of Majiju Doss, where elongated zones of higher travel time suggest constrained movement along limited pathways. Overall, the surface reveals a clear east–west gradient, with accessibility decreasing progressively away from the urban core.

7.2 School Accessibility Zones



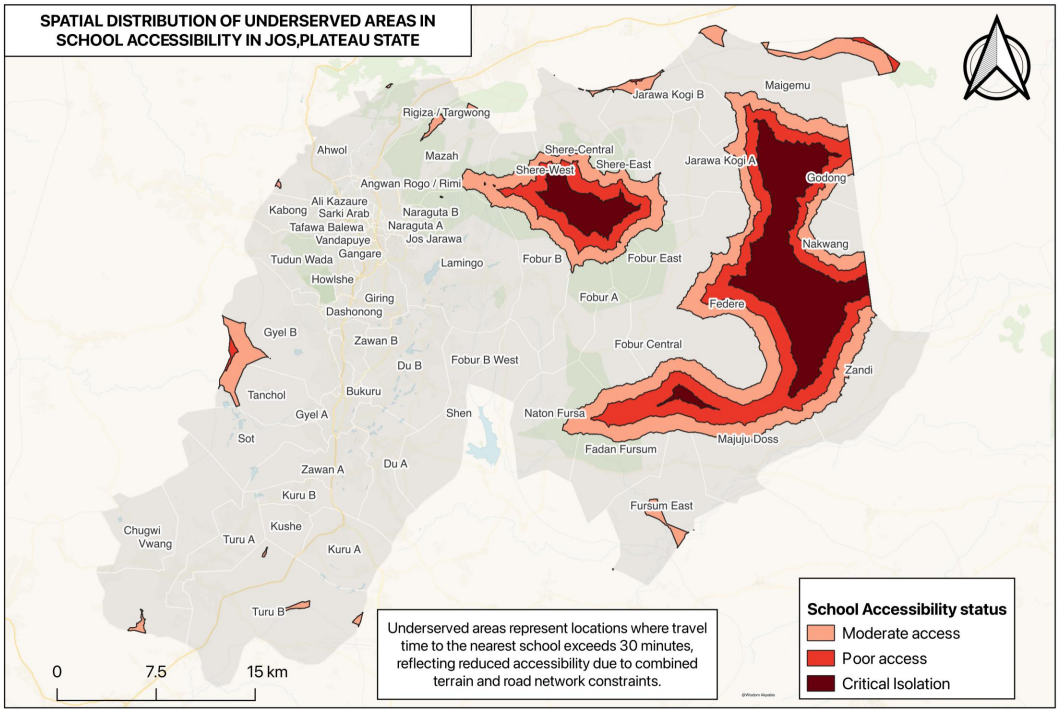
The classification of travel time into discrete accessibility zones translates the continuous surface into a more interpretable framework, highlighting spatial disparities in access to schools. The 0–15 minute zone, representing excellent accessibility, is concentrated almost entirely within the urban core. Wards such as Tudun Wada, Gangare, Angwan Rogo, Naraguta A/B, and Jos Jarawa fall within this category, reflecting both proximity to schools and strong road connectivity. These areas function as the primary accessibility hub of the study area.

Surrounding this core is a 15–30 minute transitional belt, encompassing wards such as Lamigo, Giring, and parts of Angwan Rogo's periphery. These areas retain relatively good access but begin to show increasing travel costs. Beyond this, the 30–45 minute category appears in wards such as Fobur A, Fobur B, and inner sections of Federa, marking the threshold at which accessibility begins to decline more noticeably. This transition is significant, as it represents the point where travel becomes less convenient and potentially restrictive.

The most constrained conditions are observed in the 45–60 minute and >60 minute zones, which are concentrated in peripheral wards. These include Jarawa Kogi A, Godong, Nakwang, Zandi, and outer sections of Federa, as well as parts of Fadan Fursum and Naton Fursa. In these areas, access to schools is significantly limited, reflecting both distance and movement constraints.

The spatial distribution of these zones reveals a clear hierarchical structure: a highly accessible urban core, a transitional peri-urban belt, and a peripheral zone of declining accessibility.

7.3 Spatial Distribution of Underserved Areas



The underserved areas map, defined by a travel time threshold of greater than 30 minutes, consolidates the accessibility analysis into a clear identification of priority zones. The most prominent underserved region is located in the eastern part of the study area, forming a continuous corridor across Jarawa Kogi A, Godong, Nakwang, and extending toward Zandi. This area is characterized by extensive zones of poor access and critical isolation, indicating a persistent and large-scale accessibility deficit. The continuity of this cluster suggests that the issue is structural, driven by both terrain and limited infrastructure.

A second, more localized underserved cluster is evident around Shere-West and parts of Shere-Central, where a concentrated pocket of reduced accessibility exists within an otherwise moderately accessible region. This pattern points to localized imbalances in infrastructure or school distribution. The southern fringe, particularly Fadan Fursum and Naton Fursa, forms a third underserved zone, appearing as a linear band of reduced accessibility. This configuration suggests constrained movement along limited routes, likely influenced by both terrain and sparse road connectivity.

Smaller, fragmented pockets of underserved areas are also observed along the western boundary, including parts of Tanchol, but these are spatially limited and less significant in scale. Overall, the underserved areas map reinforces the findings from the previous analyses, clearly delineating

zones where accessibility constraints are most pronounced. The eastern corridor, in particular, emerges as the most critical area of concern, both in terms of spatial extent and severity of isolation.

7.4 Population Distribution Across Accessibility Zones

Table 7.1: Distribution of Child Population Across Accessibility Zones

Accessibility Level	Travel Time (Minutes)	Child Population
Excellent Access	0–15	663,540
Good Access	15–30	5,841
Moderate Access	30–45	574
Poor Access	45–60	52
Critical Isolation	>60	31

The spatial patterns observed in the maps are further clarified when examined in relation to population distribution. The results reveal a highly uneven concentration of children across accessibility classes. An overwhelming majority of children—663,540—are located within the 0–15 minute accessibility zone. This strongly reflects the concentration of population within the urban core, particularly in wards such as Tudun Wada, Gangare, Naraguta, and Angwan Rogo, where both infrastructure and school availability are highest. This indicates a strong alignment between population distribution and educational service provision in these areas.

Beyond this range, population declines sharply. Only 5,841 children fall within the 15–30 minute category, representing a rapid transition from optimal to moderate accessibility conditions. This sharp drop highlights the extent to which both population density and service coverage diminish outside the urban core. The decline becomes even more pronounced beyond the 30-minute threshold. Only 574 children are located within the 30–45 minute range, followed by 52 children in the 45–60 minute category and just 31 children in areas of critical isolation exceeding 60 minutes. These values indicate that relatively few children reside in areas of severe accessibility constraint.

However, when considered alongside the spatial extent of underserved areas—particularly in wards such as Jarawa Kogi A, Godong, Nakwang, Zandi, and Fadan Fursum—a critical distinction emerges between spatial coverage and demographic impact. While underserved areas occupy substantial portions of the landscape, they contain only a small fraction of the total child population. This reveals a dual reality. On one hand, the current distribution of schools and infrastructure is highly efficient in serving the majority of children. On the other hand, small populations in peripheral wards experience disproportionately high barriers to access. This highlights an important trade-off between efficiency and equity in service provision.

The sharp decline in population beyond the 30-minute threshold also reinforces its relevance as a meaningful boundary for defining underserved areas, both spatially and demographically.

7.5 Synthesis of Findings

Taken together, the results demonstrate a clear and structured pattern of accessibility across Jos. The urban core functions as a zone of optimal accessibility, supported by dense infrastructure and facility distribution. Surrounding this is a transitional belt of moderate access, beyond which accessibility declines sharply in peripheral regions.

The analysis confirms that infrastructure plays a decisive role in shaping accessibility outcomes. While terrain contributes to movement constraints, its effects are significantly mitigated in areas with well-developed road networks. Conversely, in wards such as Godong, Nakwang, and Zandi, limited infrastructure amplifies terrain-related barriers, resulting in severe accessibility constraints. Overall, accessibility in Jos is characterized by strong central efficiency and peripheral inequality, with clearly defined zones of concern that provide a basis for targeted planning and intervention.

8. JUSTIFICATION OF THE STUDY

Understanding spatial accessibility to educational services is essential for achieving equitable development outcomes, particularly in rapidly urbanizing regions of the Global South. While significant attention has been given to increasing the number of schools, less emphasis has been placed on evaluating whether these facilities are effectively accessible to the populations they are intended to serve. This study is justified on conceptual, methodological, and policy grounds.

8.1 Conceptual Justification

Most conventional assessments of school access rely on simple distance-based measures, implicitly assuming uniform movement conditions across space. Such assumptions are inadequate in environments like Jos, where terrain variability and infrastructure disparities significantly influence mobility.

By adopting a cost-based modeling approach that integrates terrain and transportation networks, this study advances a more realistic understanding of accessibility. It shifts the focus from “how far” schools are to “how difficult it is to reach them,” thereby aligning accessibility analysis with real-world movement dynamics.

8.2 Methodological Justification

This study contributes methodologically by demonstrating a fully integrated GIS workflow that combines:

- Terrain-derived impedance (from DEM and slope)
- Road network friction (based on functional classification)
- Cost-distance analysis (least-cost path modeling)

- Population-based impact assessment (zonal statistics)

While each of these components has been used in isolation in previous studies, their integration into a single analytical framework enhances both accuracy and interpretability. Furthermore, the translation of spatial accessibility into population-level metrics addresses a key gap in many GIS-based studies, which often stop at mapping without quantifying impact.

5.3 Policy and Planning Justification

From a policy perspective, the study provides a decision-support framework for educational planning. By identifying not only where accessibility is limited but also how many children are affected, the analysis enables more targeted and efficient allocation of resources.

In the context of Nigeria's commitment to improving educational access under national development plans and global frameworks such as the Sustainable Development Goals (SDGs), this study offers a practical tool for:

- Identifying underserved areas
- Prioritizing infrastructure development
- Supporting equitable service delivery

5.4 Local Relevance

Jos represents a rapidly evolving urban landscape with a mix of dense urban cores and expanding peri-urban settlements. The coexistence of complex terrain and uneven infrastructure distribution makes it particularly suitable for advanced accessibility modeling.

The insights generated from this study are therefore not only academically relevant but also directly applicable to local planning contexts, where decisions must account for both physical and demographic realities.

9. LIMITATIONS OF THE STUDY

Despite the robustness of the analytical framework, several limitations must be acknowledged. These limitations relate primarily to data quality, modeling assumptions, and the inherent complexity of human mobility.

9.1 Population Data Representation

The population dataset used in this study is derived from gridded raster data, which, while useful for spatial analysis, may not fully capture the distribution of population in rural or informal settlements. In particular, sparsely populated areas at the urban fringe may be underrepresented, leading to potential underestimation of the number of children affected by poor accessibility.

This limitation is reflected in the extremely low population counts observed in high travel-time zones, which may not entirely correspond to on-ground realities.

9.2 Assumptions in Travel Speed and Friction Values

Road friction values were assigned based on generalized assumptions about travel speeds associated with different road types. These assumptions do not account for:

- Variations in road condition (e.g., paved vs unpaved)
- Traffic congestion in urban areas
- Seasonal factors affecting road usability

Similarly, the conversion of slope into travel time is based on simplified relationships that may not fully represent human movement behavior under varying conditions.

9.3 Static Nature of the Model

The model represents a static snapshot of accessibility and does not incorporate temporal variability. In reality, accessibility can change significantly over time due to:

- Weather conditions (e.g., rainfall affecting road conditions)
- Time-of-day variations (e.g., peak traffic periods)
- Infrastructure changes

The absence of temporal dynamics limits the ability of the model to capture these fluctuations.

9.4 Simplification of Movement Behavior

Human movement is influenced by a range of factors beyond terrain and roads, including:

- Mode of transport (walking, cycling, vehicular)
- Socioeconomic constraints
- Cultural and behavioral preferences

The model assumes a generalized movement behavior and does not differentiate between these factors, which may lead to oversimplification of accessibility patterns.

9.5 Data Completeness and Accuracy

The road network dataset, derived from OpenStreetMap, may contain inconsistencies or omissions, particularly in less mapped areas. Missing or inaccurately classified roads can affect the reliability of the road friction surface and, by extension, the integrated cost surface.

9.6 Exclusion of Facility Capacity and Quality

The analysis treats all schools as equal in terms of service provision, without accounting for differences in:

- Capacity (number of students accommodated)
- Quality of facilities
- Availability of resources

As a result, the model evaluates physical accessibility only, without considering whether schools can adequately serve the populations within their catchment areas.

10. CONCLUSION

This study developed a spatially explicit framework for evaluating school accessibility in Jos by integrating terrain and road network characteristics into a unified cost-distance model. By moving beyond traditional distance-based approaches, the analysis provides a more realistic representation of movement across space and reveals how accessibility is shaped by the interaction of physical geography, infrastructure, and settlement patterns.

The results demonstrate that accessibility in Jos is highly structured and spatially uneven. The urban core, particularly wards such as Tudun Wada, Gangare, Naraguta, and Angwan Rogo, functions as a zone of optimal accessibility. In these areas, dense road networks and a high concentration of educational facilities ensure that the majority of children can reach schools within short travel times. This indicates that, from an efficiency perspective, the current distribution of schools is well aligned with population concentration.

However, the analysis also reveals clear zones of reduced accessibility in the peripheral regions of the study area. The eastern corridor, spanning Jarawa Kogi A, Godong, Nakwang, and Zandi, emerges as the most significant area of concern, characterized by extensive travel times and continuous spatial isolation. Similarly, the southern fringe, including Fadan Fursum and Naton Fursa, exhibits constrained accessibility due to limited infrastructure and terrain-related barriers. Although the population within these areas is relatively small, the severity of the accessibility constraints highlights a persistent form of spatial inequality.

A key finding of the study is the existence of a distinct threshold at approximately 30 minutes of travel time. Within this range, accessibility remains relatively stable and favorable, but beyond it, conditions deteriorate rapidly. This threshold provides a meaningful and practical basis for distinguishing between adequately served and underserved areas.

The study further demonstrates that infrastructure plays a decisive role in shaping accessibility outcomes. While terrain contributes to movement constraints, its effects are significantly reduced in areas with well-developed road networks. In contrast, in wards where infrastructure is limited, terrain effects are amplified, leading to disproportionately high travel times. This underscores the importance of connectivity as a critical determinant of access.

Overall, the findings indicate that while the majority of children in Jos benefit from favorable access to schools, small but spatially significant populations remain underserved. These disparities reflect not only differences in distance but also variations in movement conditions, highlighting the need for spatially informed planning approaches.

11. RECOMMENDATIONS

Improving school accessibility in Jos requires a targeted and context-sensitive approach that addresses both infrastructure deficits and spatial inequalities. Priority should be given to enhancing road connectivity in underserved regions, particularly along the eastern corridor encompassing Jarawa Kogi A, Godong, Nakwang, and Zandi, as well as the southern fringe areas such as Fadan Fursum and Naton Fursa. Strengthening road networks in these locations would significantly reduce travel times and improve access to existing educational facilities without necessarily requiring extensive new construction.

At the same time, the strategic placement of additional schools should be considered in areas where accessibility remains critically low and where identifiable population clusters exist. Such interventions should be guided by the 30-minute accessibility threshold identified in this study, ensuring that new facilities are located in areas where they can have the greatest impact on reducing travel time and improving service coverage.

Future analyses should incorporate higher-resolution and locally validated population datasets to better capture the distribution of children in rural and peri-urban areas. This would improve the accuracy of accessibility assessments and ensure that underserved populations are not underestimated. Additionally, incorporating multiple modes of transport and accounting for temporal variations in travel conditions would enhance the realism of the model and provide a more comprehensive understanding of accessibility dynamics.

There is also a need to institutionalize geospatial accessibility modeling within planning processes. By adopting GIS-based cost-distance frameworks as standard tools, planning authorities can make more informed decisions regarding the allocation of educational resources and the prioritization of infrastructure development.

Ultimately, improving accessibility requires a shift from generalized planning approaches to spatially targeted interventions that recognize the uneven geography of access. By focusing on areas where both movement constraints and population needs intersect, policymakers can achieve a more equitable distribution of educational services across Jos.